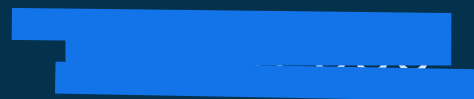


SIP Presentation

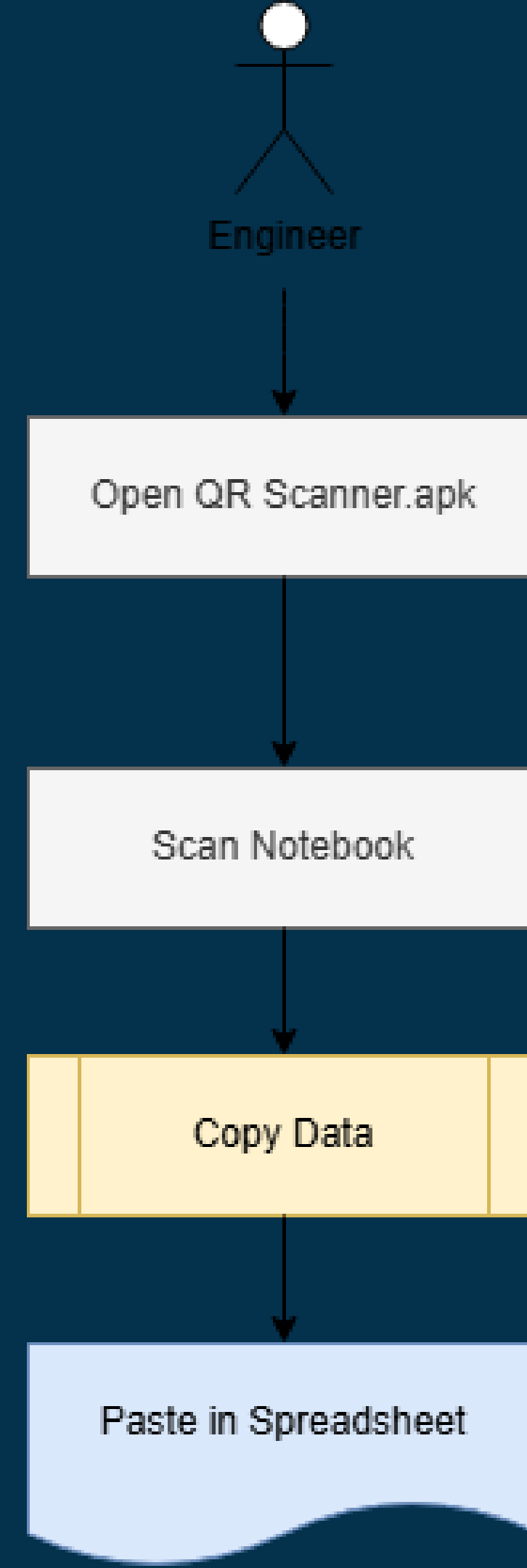
QR-Based Deployment Tracking System

Irfan Aiman bin Mohd Yussuf



Background of Study

- Involvement in large-scale IT device deployment operations
- Existing workflow relied on manual data entry into Excel spreadsheets
- Increasing deployment volume created operational bottlenecks
- High dependency on manual processes impacted:
 1. Operational Efficiency
 2. Data Integrity
 3. Process Scalability



Problem Statements



LACK OF REAL TIME VISIBILITY

- No immediate tracking of deployment progress



INEFFICIENT REPORTING PROCESS

- Manual data cleaning and consolidation required



DATA INACCURACY RISKS

- Human errors in manual input
- Duplicate and inconsistent records

LIMITED SCALABILITY

- System unable to support large-scale operations effectively





OBJECTIVE OF STUDY



DEVELOP AN AUTOMATED DEPLOYMENT TRACKING SYSTEM



IMPLEMENT QR-BASED DATA CAPTURE MECHANISM



ESTABLISH A CENTRALIZED CLOUD DATABASE

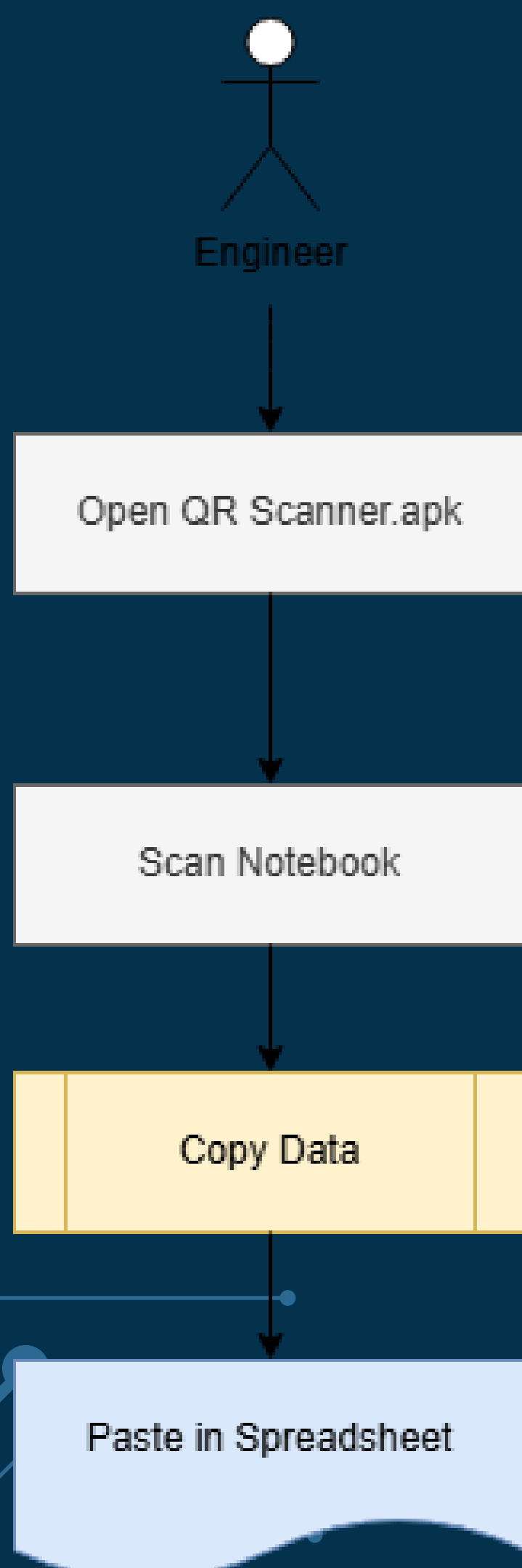


ENHANCE DATA INTEGRITY THROUGH VALIDATION MECHANISMS

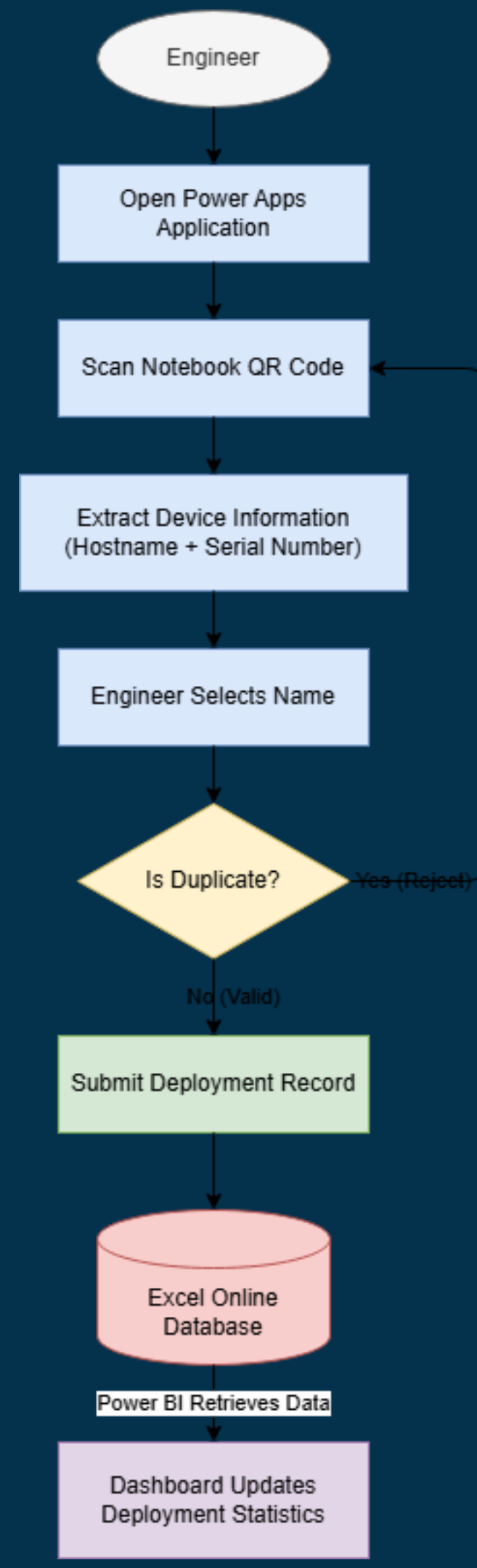


ENABLE REAL-TIME ANALYTICS USING POWER BI

Traditional



Modern



SCOPE OF STUDY

Application Layer

Development of mobile application using Microsoft Power Apps

Data Management Layer

Centralized storage using Excel on OneDrive

Analytics Layer

Visualization using Microsoft Power BI dashboards

Literature Review – 1

QR CODE TECHNOLOGY IN DEVICE TRACKING

- 2-Dimensional Barcode that can store a variety of data types
- Can store Hostname & Serial Number
- Reduce time taken and minimize human errors
- Cost effectiveness and ease to implement.
- Most smartphones and tablets already support QR scanning through built in camera functions or applications.

LOW-CODE DEVELOPMENT PLATFORMS

- The use of low-code helped to address software development
- Microsoft Power Apps is one of the the most widely adopted low-code platforms.
- Quick and fast development of applications.

Literature Review – 2

CLOUD-BASED DATA STORAGE SYSTEMS

- Cloud computing has emerged as an integral part of modern Information System.
- OneDrive is one of the most commonly used as the cloud storage.
- Cloud-based information provide accessibility.
- In this context of project, Microsoft Excel stored in OneDrive was used as the centralized database for storing notebook deployment records.

BUSINESS INTELLIGENCE AND DATA VISUALIZATION

- BI was used for collecting, analyzing, and presenting information or data.
- Transforming raw data into structured information.
- Helps managements to obtain real time information about operational activities
- Identify areas that need improvements.
- In this project, Microsoft Power Bi was used to provide information about notebook deployment activities

Methodology

SYSTEM ARCHITECTURE OVERVIEW

Engineer (Mobile Device)

Power Apps Application
(QR Code Scanning + Data Validation)

Excel Online Database (OneDrive)
(Centralized Deployment Records)

Power BI Dashboard
(Deployment Analytics & Monitoring)

APPLICATION LAYER

- Developed using Microsoft Power Apps.
- Engineers can use to scan QR Code using their camera.
- Engineers also required to select their names before submit to database.

STORAGE LAYER

- Used as database to store deployment record.
- Developed using Microsoft Excel and OneDrive.
- Excel table contains hostname, serial number, engineer name, timestamp.
- Securely stored in cloud environment and can be accessed by other applications.

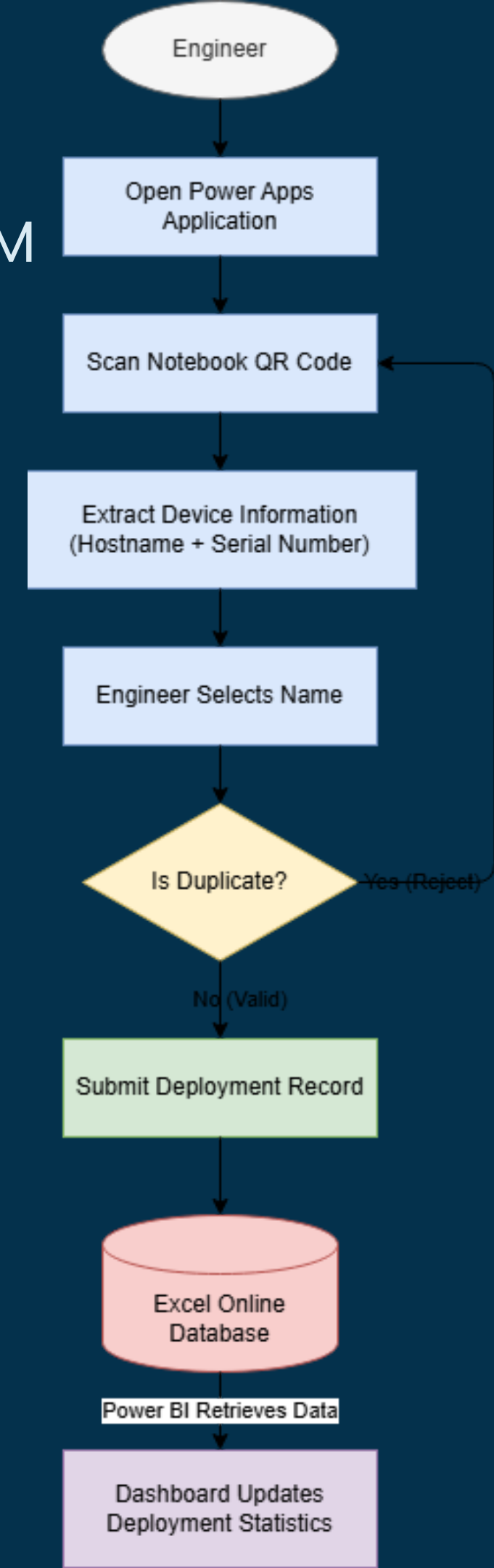
VISUALIZATION LAYER

- Developed using Microsoft Power BI.
- Transform raw data to informative dashboard.
- Enables management to monitor deployment activities and make informed decision.

Methodology

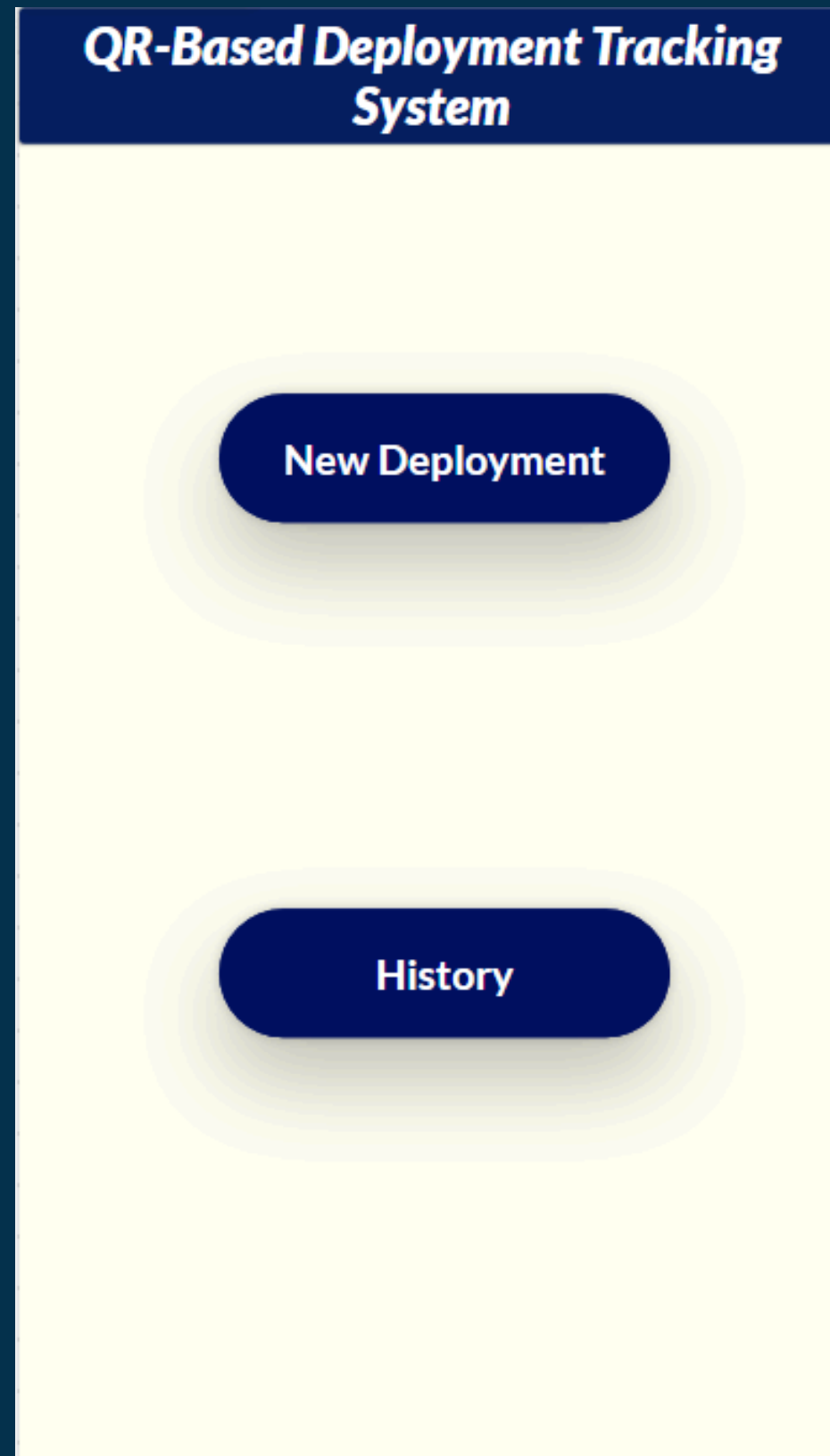
FLOW OF DEPLOYMENT TRACKING SYSTEM

- Launch the Power Apps mobile application.
- Each Device has a unique identifier in the form of QR Code
- Engineers scan the QR Code and retrieve the information contained in the code
- Engineer can select their name from the dropdown list.
- Validation process ensures that a device has not been previously recorded in the database.
- Submission of the validated deployment record into centralized database.
- Then a query is sent by the POWER BI to retrieve the updated data from excel database.
- The Processed data is then used to create a dashboard.



Methodology

APPLICATION DEVELOPMENT (USER INTERFACE)



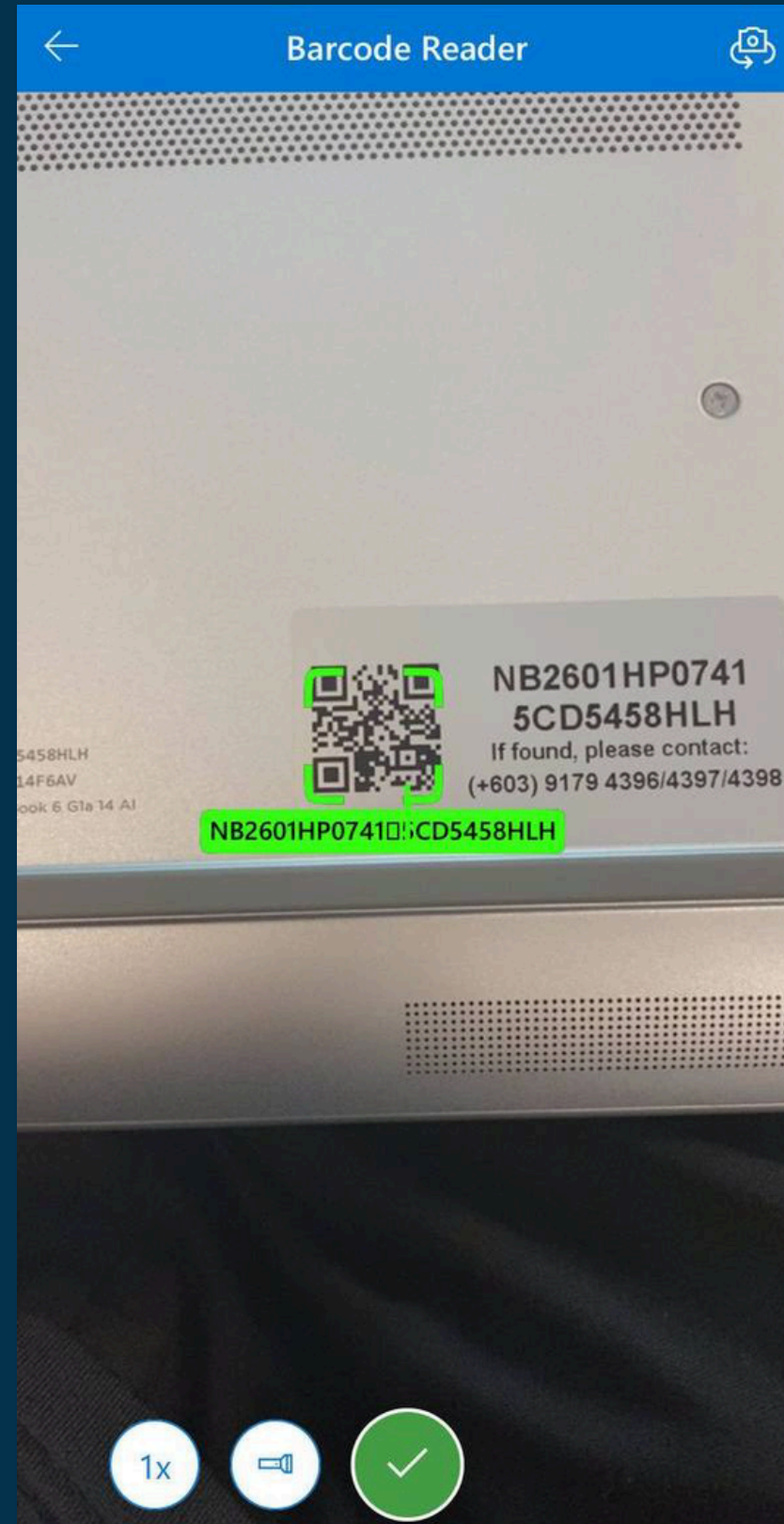
1.1 Main Page



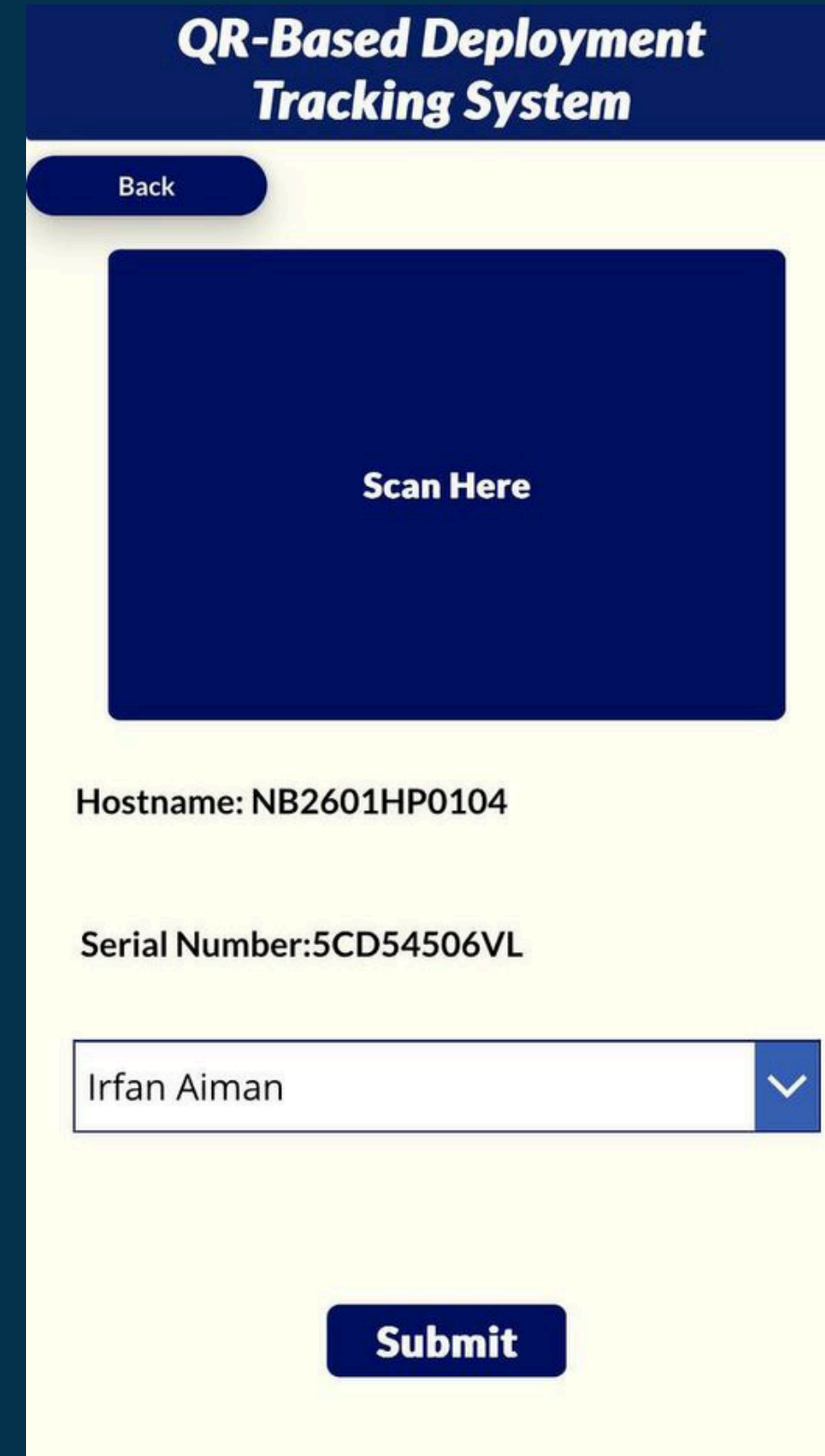
1.2 History Page

Methodology

APPLICATION DEVELOPMENT (QR CODE + DATA SUBMISSION)



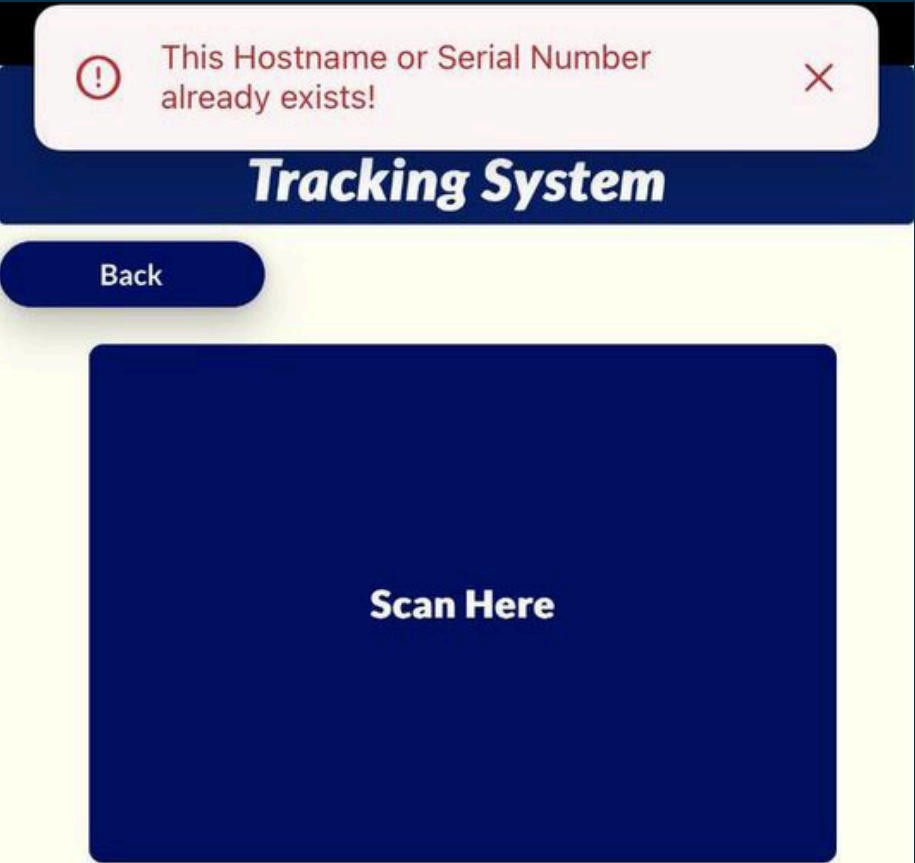
1.3 QR Code Scanner



1.4 Deployment Data Entry Interface

Methodology

APPLICATION DEVELOPMENT (DUPLICATE RECORD VALIDATION + TIMESTAMP)



1.5 Duplicate Record Notification

```
!IsBlank(  
  LookUp(  
    Scan_Data_table,  
    Hostname = varHostText || Serial_Number = varSerialText  
  ),  
,  
Notify(  
  "This Hostname or Serial Number already exists!",  
  NotificationType.Error  
)  
)
```

1.6 Duplicate Record Detection on Submit Button

26/1/2026 11:55
26/1/2026 14:27
26/1/2026 15:27
27/1/2026 9:14
27/1/2026 10:10
27/1/2026 10:11
27/1/2026 10:11
27/1/2026 10:12
27/1/2026 10:23
27/1/2026 11:20
27/1/2026 11:22
27/1/2026 11:32
27/1/2026 14:14
27/1/2026 15:22
27/1/2026 15:31
27/1/2026 16:54
27/1/2026 16:55
28/1/2026 9:36
28/1/2026 9:37
28/1/2026 9:38
28/1/2026 9:38
28/1/2026 9:39

1.7 Deployment Timestamp Records

Methodology

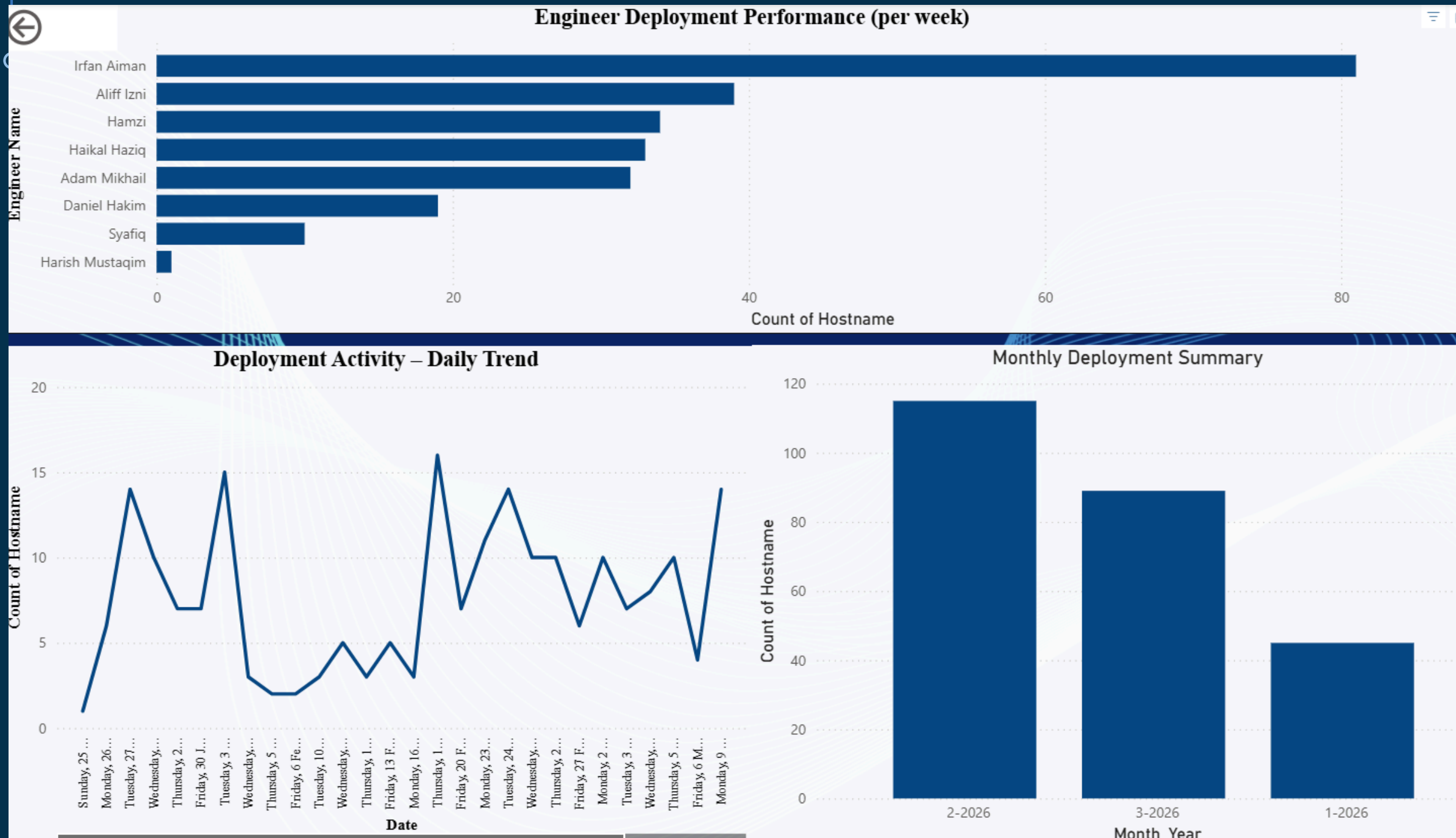
DATA CLEANING AND TRANSFORMATION

- The raw data has to be cleaned and transformed
- Timestamp from deployment data need to be converted to a DateTime format.
- "Scan_Date_Only" and "Month_Year" are created from the original timestamp.

Scan_Date_Only	Month_Year
Sunday, 25 January, 2026	1-2026
Monday, 26 January, 2026	1-2026
Monday, 26 January, 2026	1-2026
Monday, 26 January, 2026	1-2026
Monday, 26 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Tuesday, 27 January, 2026	1-2026
Wednesday, 28 January, 2026	1-2026
Wednesday, 28 January, 2026	1-2026
Wednesday, 28 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Thursday, 29 January, 2026	1-2026
Friday, 30 January, 2026	1-2026
Friday, 30 January, 2026	1-2026
Friday, 30 January, 2026	1-2026
Friday, 30 January, 2026	1-2026
Tuesday, 3 February, 2026	2-2026
Tuesday, 3 February, 2026	2-2026
Tuesday, 3 February, 2026	2-2026
Tuesday, 3 February, 2026	2-2026
Wednesday, 4 February, 2026	2-2026
Wednesday, 4 February, 2026	2-2026
Wednesday, 4 February, 2026	2-2026
Thursday, 5 February, 2026	2-2026
Thursday, 5 February, 2026	2-2026
Friday, 6 February, 2026	2-2026
Friday, 6 February, 2026	2-2026

Methodology

DEPLOYMENT ANALYTICS VISUALIZATION

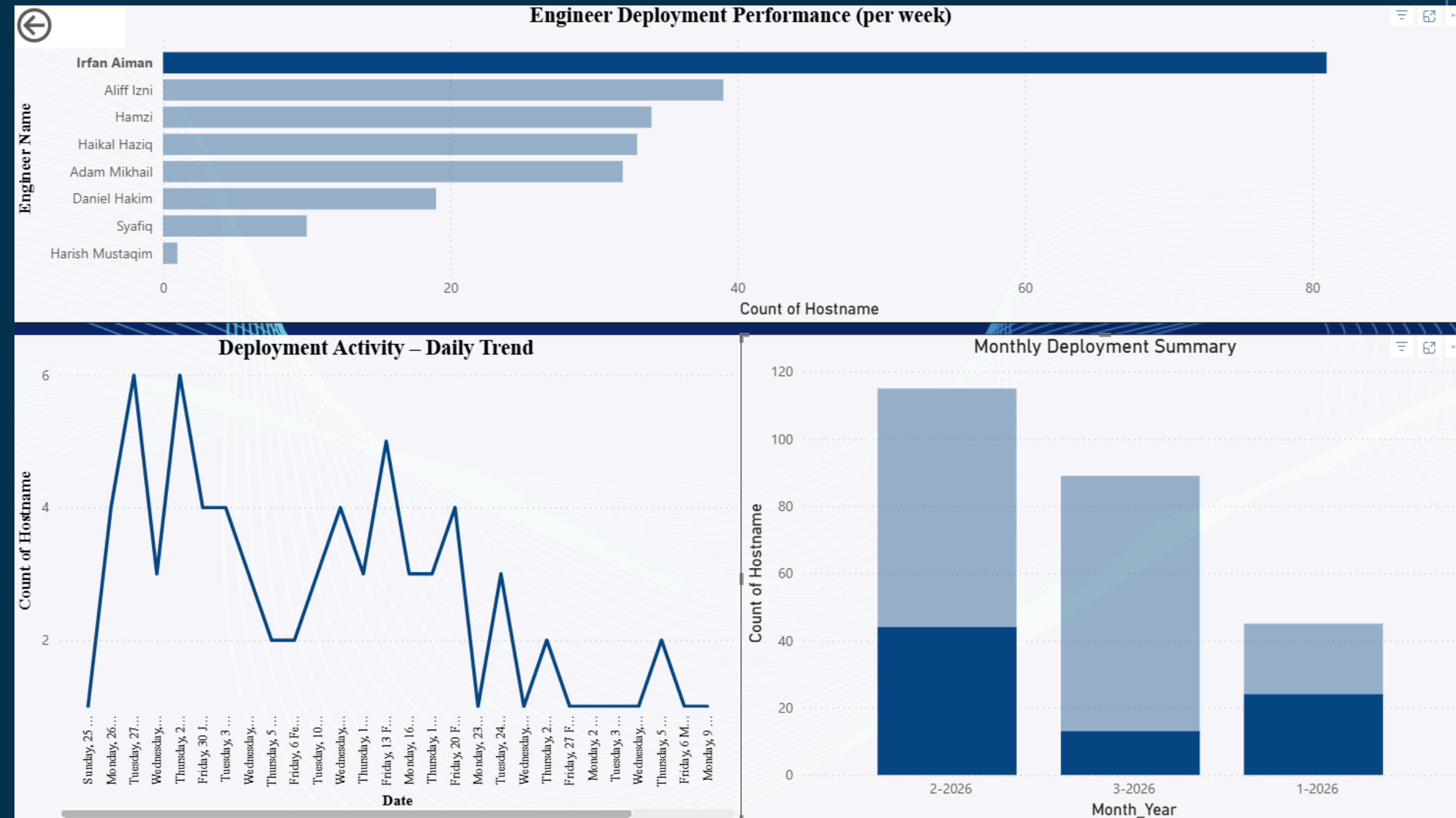


- Engineer Performance Chart : to obtain insights regarding the number of notebooks deployed by each engineer.
- Daily Deployment Chart : allows the management to gain insights regarding the number of devices deployed per day.
- Monthly Deployment Chart : obtain insights into the total number of devices deployed in a particular month.
- Using Power BI dashboard makes complex data easier to interpret and can obtain insights into overall performance of the team.

Methodology

INTERACTIVE DASHBOARD FILTERING

- Interactive features deployed data to be filter in a particular way.
- A slicer was added to improve the interactivity of the Power BI.
- When a specific engineer is selected from the slicer, the entire visualizations on the dashboard automatically change to show the deployment made by the specific engineer.



RESULT & DISCUSSION – 1

SYSTEM IMPLEMENTATION RESULTS

- The implementation was a **SUCCESS**.
- Fully functional end-to-end workflow.
- Able to provide real-time insights on the performance of the deployment.
- Also, the proposed system is a replacement of the manual entry of the information into the spreadsheet.
- The Integration of different components was a success.
- This also shows that the system is able to support real-world operation.

POWER APPS FUNCTIONALITY TESTING

- Functionality of the QR Code scanner was tested by scanning various notebook QR codes.
- Submission feature was carried out by submitting several deployment records through the application.
- Duplicate validation feature was tested by attempting to submit the same QR code.
- Timestamp recording feature was tested to ensure every record includes the timestamp information.
- Result: it is clear that the application is functioning **CORRECTLY**

RESULT & DISCUSSION – 2

DISCUSSION OF FINDINGS

- QR-Based Deployment tracking system was a **SUCCESS** providing a solution to manual data entry.
- Simple and easy to use.
- Enables the engineers to easily record the activities of the deployments without any difficulties.
- Provide real-information to the management of the firm through the use of Power BI.

DISCUSSION ON SUSTAINABILITY

Supports Digital
Transformation
Initiatives

Reduces Manual
Paper-Based
Processes

Enables Scalable
System
Enhancement

Promotes
Sustainable IT
Operations

CONCLUSION & RECOMMENDATIONS

The QR-Based Deployment Tracking System **successfully** demonstrates the application of low-code technologies to automate IT deployment processes. By integrating Microsoft Power Apps, Excel (OneDrive), and Power BI, the system improves operational efficiency, data accuracy, and real-time visibility while eliminating manual recording and reducing errors. Validation mechanisms ensure data integrity, and dashboards support data-driven decision-making. Future enhancements, including scalable data storage, user authentication, and advanced analytics, further strengthen its potential as a comprehensive and scalable digital solution aligned with digital transformation initiatives.



Thank
You



